



Aduvani Vanjari Akanksh

Center for Machine Intelligence and Data Science

M.S. by Research (CMInDS), IIT Bombay

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ACADEMIC QUALIFICATIONS

Program	Discipline	University	Year	GPA
MS by Research (MSR)	Artificial Intelligence and Data Science	IIT Bombay	2024 - Present	9.15/10
Bachelor of Technology	Electrical Engineering	IIT Tirupati	2018 - 2022	8.63/10
Intermediate Education (XII)	MPC	TSBIE	2016 - 2018	95.5%
Secondary Education (X)	MPC	TSBSE	2016	9.5/10

PROFESSIONAL EXPERIENCE

Software Developer (Backend) at Paytm

Jan'23 - Dec'23

FSM Team & BINARY Team

Paytm, India

- Implemented new features and enhancements to the existing code base of **Field Service Management (FSM)**
- Automated file processing related to merchant's invoice/bill generation, previously handled manually by the Revenue Assurance team

Jr. Data Scientist at Holmusk

June'22 - Aug'22

Guide: Shivshankar Umashankar

Holmusk, Singapore

- Improved the code base of **Neuroblu** web application for healthcare analytics and patient outcome prediction
- Optimized dashboard performance, reducing latency from 20 seconds to 2 seconds (**10x improvement**)

TECHNICAL SKILLS

Programming Languages	Python, C++, C, JavaScript, Java
Frameworks & Libraries	TensorFlow, PyTorch, Scikit-Learn, NumPy, SciPy, Django, React.js, Spring Boot
Tools & Technologies	Git, L ^A T _E X, MATLAB, VHDL, Model-sim, HTML, p5.js, Processing
Hardware	Arduino, Raspberry Pi
Others	InkScape

KEY PROJECTS

Option Pricing and Implied Volatility Forecasting

Jul'25 - Present

Guide: Dr. Piyush Pandey

IIT Bombay

- Performed a focused literature review of Agent-Based Models (ABM) and Large Market Models (LMM — Large Market Model, analogous to LLM) based on papers ; read and synthesized modeling choices, calibration methods and evaluation metrics.
- Summarized common ABM design patterns from the papers: heterogeneous agents (fundamentalists, chartists, liquidity providers), simple order-matching rules, and mechanisms that reproduce stylized facts (fat tails, volatility clustering).
- Currently implementing the Kronos paper as a prototype and evaluating where structural or algorithmic improvements could be introduced.

Interpreting Emotion Understanding in LLMs

Oct'25 - Nov'25

Course Project (CS 772: Deep Learning)

IIT Bombay

- Conducted mechanistic interpretability analysis on **Qwen2.5 models (0.5B-7B)** to decode how LLMs infer emotion, moving beyond black-box predictions to internal state analysis.
- Implemented **layer-wise linear probes**, discovering that emotional semantic concepts are distinct from linguistic features and localize specifically in mid-to-late transformer layers (14-23).
- Developed novel **Causal Tracing** techniques using noise ablation and activation patching to identify specific tokens and neurons responsible for triggering emotion classification.
- Introduced **"Emotion Persistence"** metrics (Half-Life, AUC) to quantify context decay and built an interactive **Streamlit dashboard** for real-time visualization of causal impacts.

Param-1 Indic-Scale Bilingual Foundation Model

Jan'25 - May'25

Guide: Kundeshwar Pundalik [AIKOSH Model details](#)

IIT Bombay

- Worked on the data curation pipeline, processed around 12TB of data using Nvidia's Nemo Framework, worked on quality filter, classification filter on datasets like Zyda, FineWeb, DCLM, and Dolma.
- Created a script to scrape data from archive.org which can be used for training the model.

Fruit Quality Estimation using Deep Learning

Guide: Dr. Rama Krishna Sai Gorthi

Jul'21 - May'22

IIT Tirupati

- Developed a **computer vision system** to estimate fruit size from depth maps extracted from stereo images
- Used **Mask RCNN architecture** along with standard ML Techniques like **Hough circles** with custom modifications for precise size estimation with less than 5% error
- Processed over 1000 fruit images to create a comprehensive dataset for training and validation
- Achieved **92% accuracy** in fruit size classification over the Pomegranate fruit

SURF: Speeded Up Robust Features

Guide: Dr. Subramanyam Gorthi

Mar'21 - May'21

IIT Tirupati

- Implemented **SURF algorithm** from scratch using Python, creating a scale and rotation invariant feature detector
- Optimized performance using integral images and Hessian matrix approximation for faster computation
- Applied the implementation for object recognition tasks, achieving **87% matching accuracy** on benchmark datasets
- Compared performance with SIFT and ORB algorithms, demonstrating SURF's computational efficiency advantages

Indoor Navigation System (SIH2020)

Self Project

Jan'20 - Feb'20

Smart India Hackathon 2020

- Designed a web application for indoor navigation in areas where GPS lacks accuracy
- Implemented **A* and Dijkstra's algorithms** for calculating optimal paths within building floor plans
- Created a user-friendly interface allowing users to upload building layouts and select waypoints
- Developed a real-time tracking system using Wi-Fi triangulation for accurate indoor positioning

POSITIONS OF RESPONSIBILITY

System Administrator

Indian Institute of Technology Bombay

Jul'24 - Dec'25

Bombay, India

- Maintain the CMInDS website and automatically backup the website when there are new changes to the website.
- Make new services for Department like OSTicket, Interview Portal, Research Publication Website, etc.

Placement Representative

Indian Institute of Technology Tirupati

Jul'21 - Jun'22

Tirupati, India

- Facilitated smooth placement processes for various organizations visiting IIT Tirupati for over 170 students
- Coordinated between companies, placement cell, and students to streamline the recruitment process
- Organized pre-placement talks, interviews, and follow-up communications with industry partners

NSS Co-coordinator

National Service Scheme, IIT Tirupati

Aug'19 - Aug'20

Tirupati, India

- Led and organized various NSS activities in nearby villages focusing on sustainable development initiatives
- Encouraged student participation in community service, achieving **80% active involvement** from the batch
- Coordinated educational outreach programs benefiting over 200 school children in rural areas

ACHIEVEMENTS

- Finished in **top 10** in the QuantQuest competition conducted by **QRT** at **IIT Bombay** among 200+ teams. (2024)
- Secured **2nd place** in "F.I.R.E" (Flame Imaging Robot Extinguisher) technical event at Tirutsava (2020)
- Secured **2nd place** in "ArduiKnow Challenge" competition during Tirutsava 2020 (2020)
- Secured **3rd place** in the Terrace Farming Robot competition at **Inter IIT Tech Meet 8.0** among 23 IITs (2019)
- Won **1st place** in "The Line Following Robot" competition held by TechManiacs club, IIT Tirupati (2019)

RELEVANT COURSEWORK

Machine Learning

Fundamentals of Machine Learning, Computer Vision, Deep Learning Techniques

Signal Processing

Signals and Systems, Digital Signal Processing, Image Processing Fundamentals

Computer Science

Data Structures and Algorithms, Digital Systems, Introduction to Robotics

Mathematics

Linear Algebra, Probability and Statistics, Optimization Techniques

ADDITIONAL PROJECTS

- **Terrace Farming Robot:** Built an autonomous robot capable of ploughing, seed sowing, watering, and harvesting, with the ability to navigate between different terrace levels (2019)
- **JPEG Compression:** Implemented the JPEG compression algorithm from scratch in a team of four, achieving compression ratios comparable to standard implementations (2019)
- **Spectrogram and Periodogram Analysis:** Generated and analyzed spectrograms and periodograms of various signals using MATLAB, applying window functions for improved frequency resolution (2020)

INTERESTS & HOBBIES

- **Personal:** Badminton, Drawing, Table Tennis, Brand Logo Designing